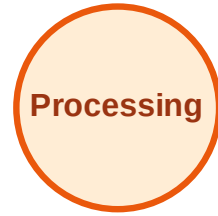


BFS Traversal

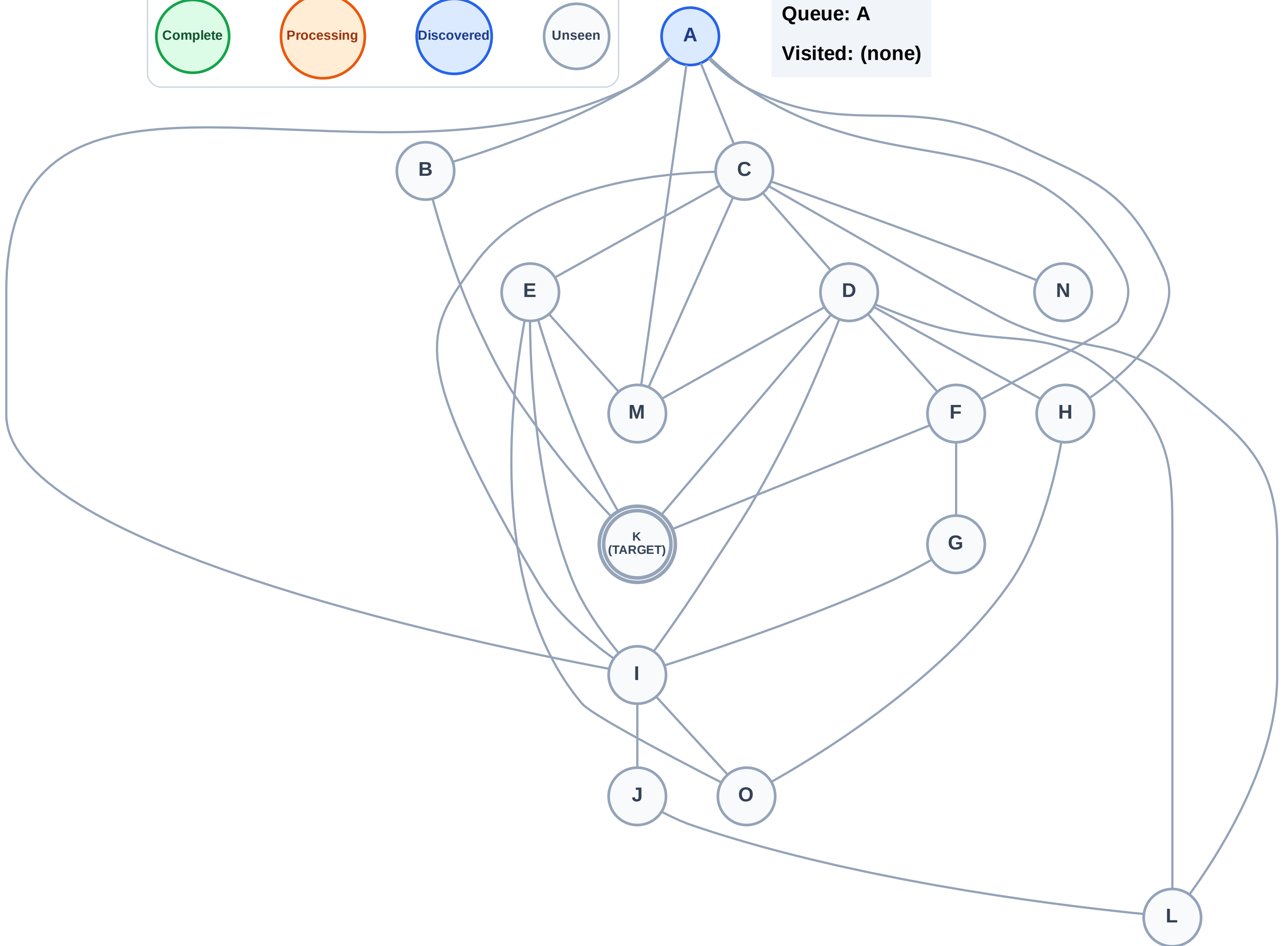
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

Complete

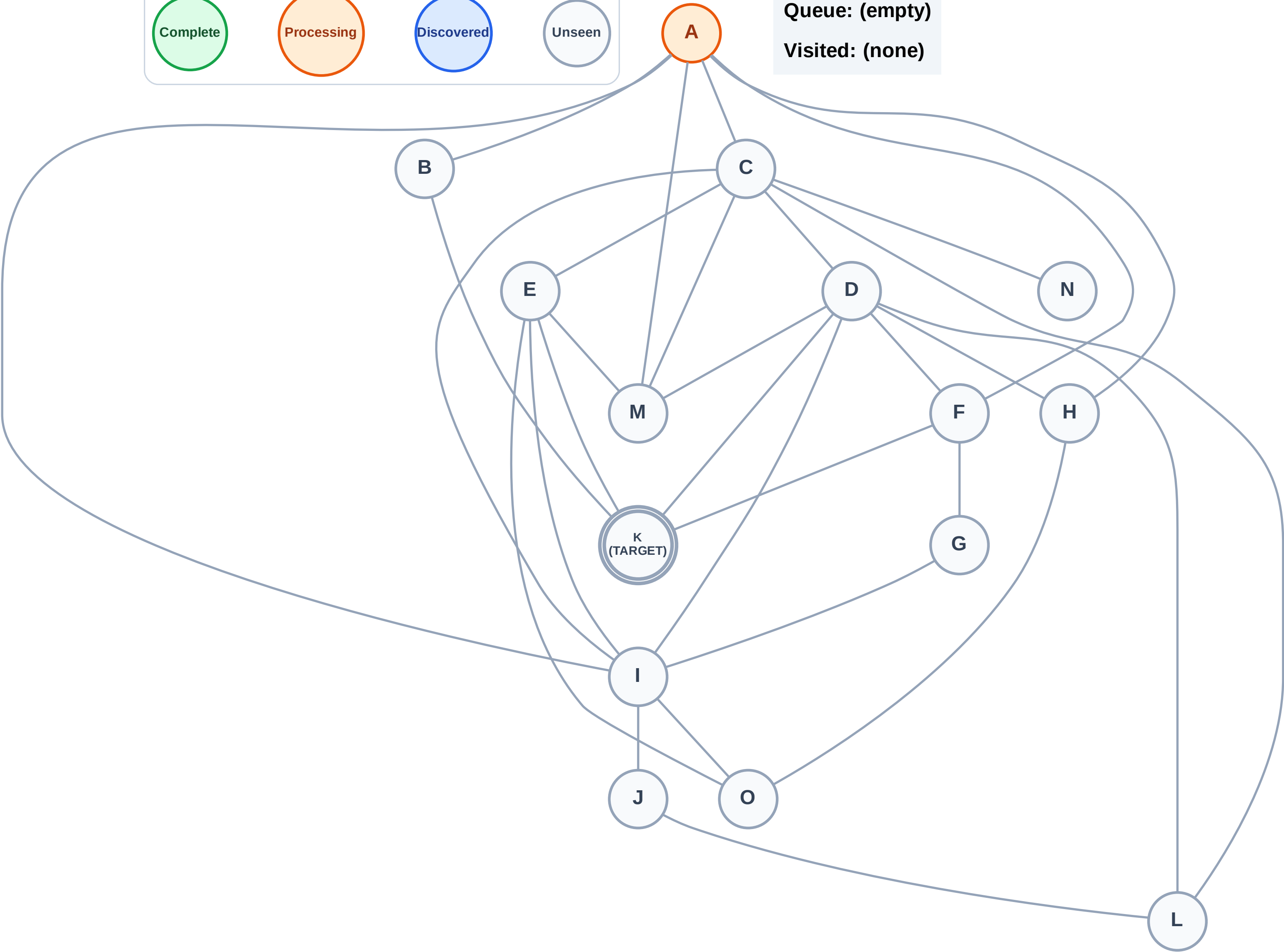
Processing

Discovered

Unseen

Queue: (empty)

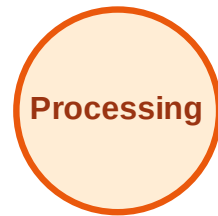
Visited: (none)



BFS Traversal

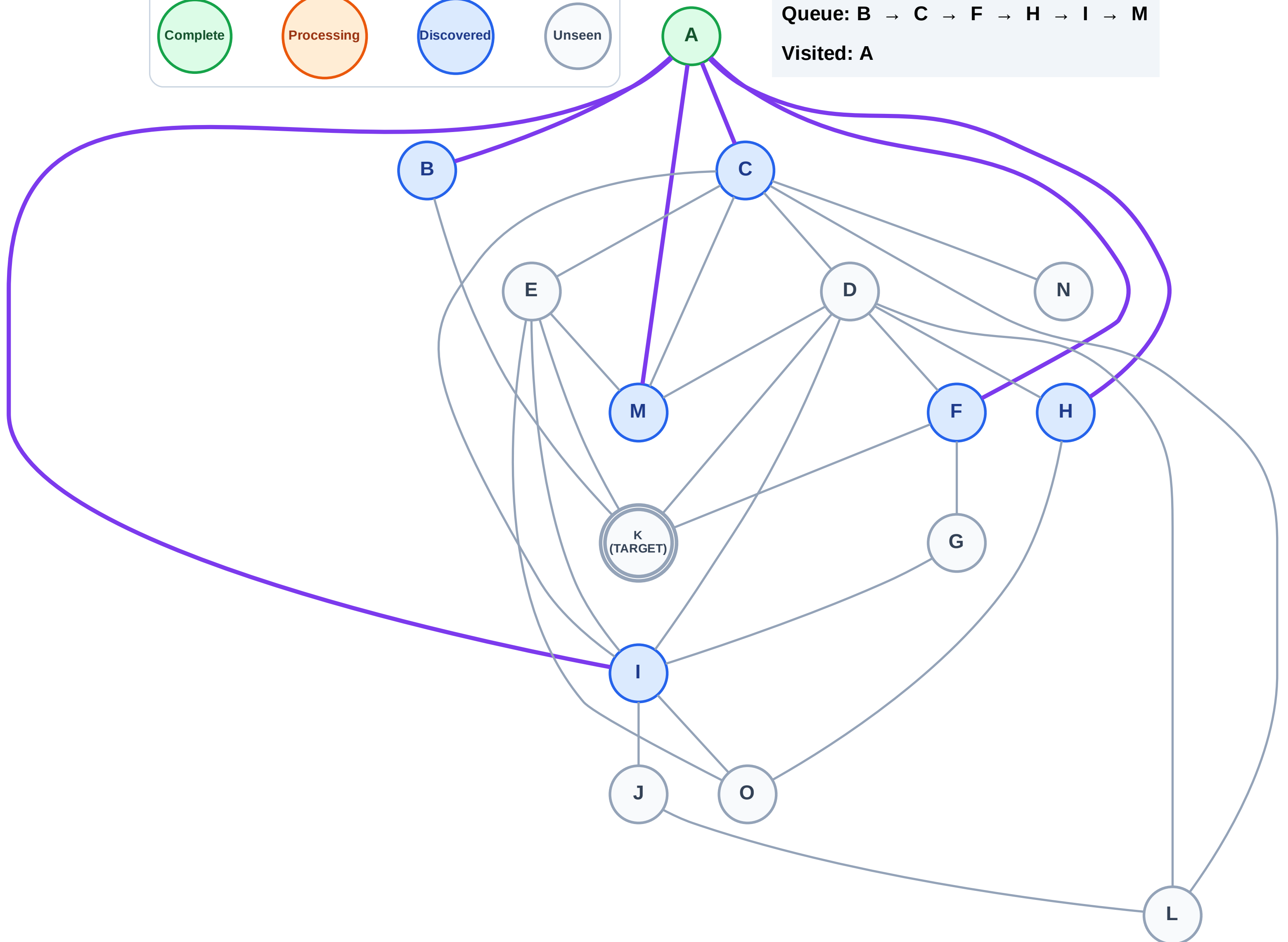
Step 3: Discover B, C, F, H, I, M from A

Legend



Queue: B → C → F → H → I → M

Visited: A



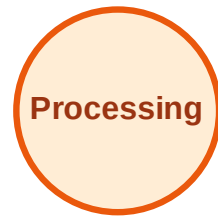
BFS Traversal

Step 4: Process node B

Legend



Complete



Processing



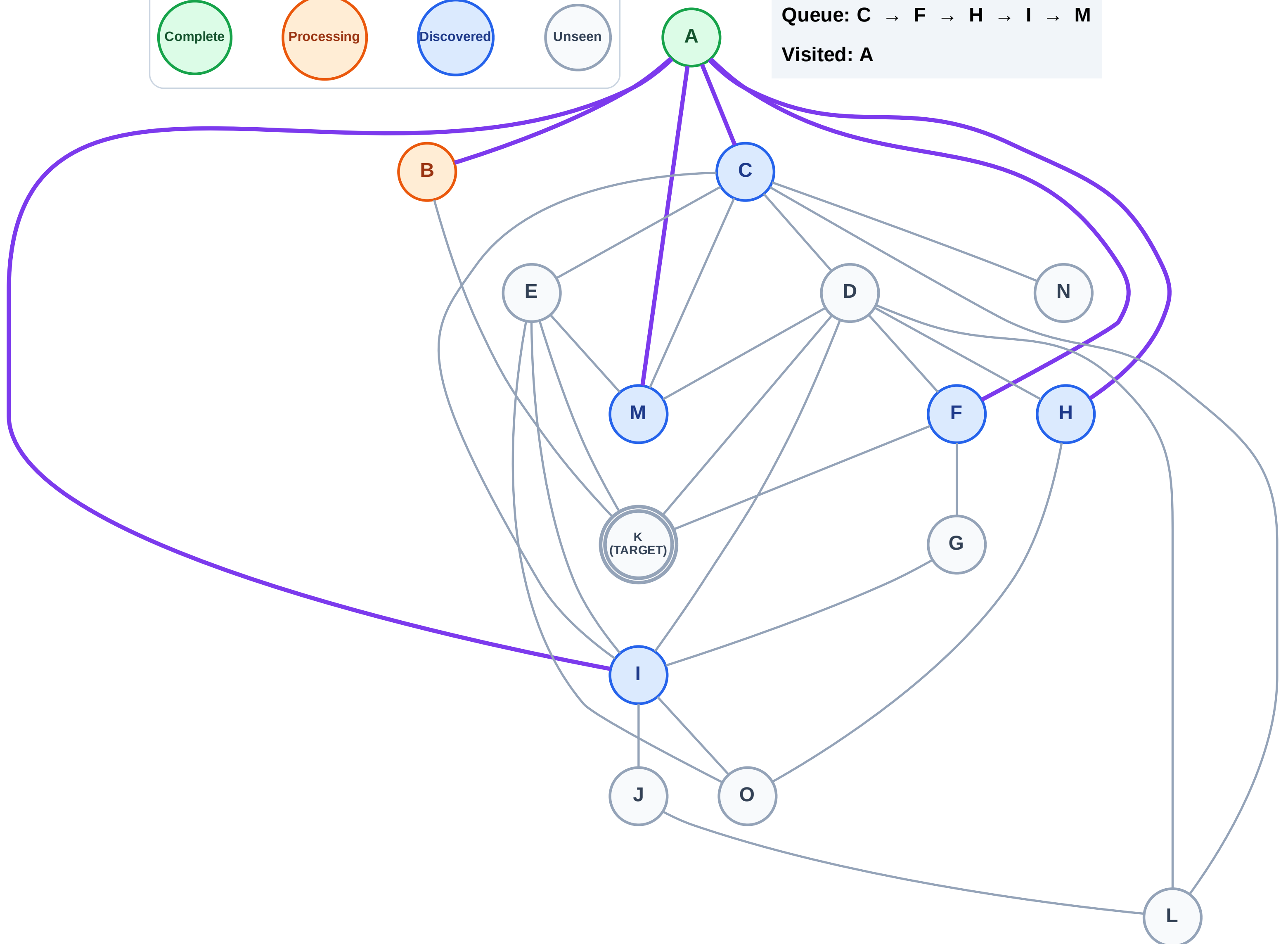
Discovered



Unseen

Queue: C → F → H → I → M

Visited: A



BFS Traversal

Step 5: Discover K from B

Legend

Complete

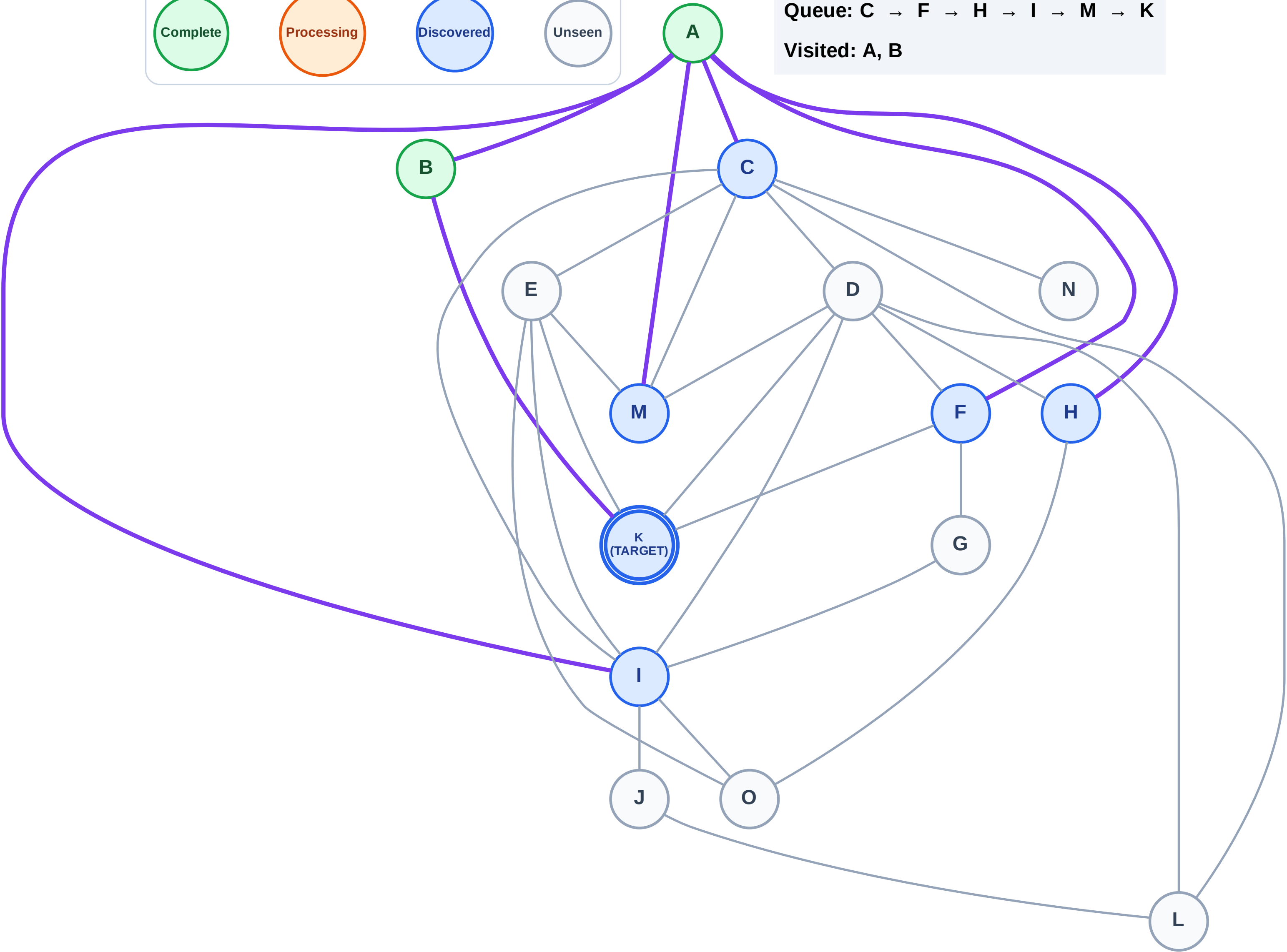
Processing

Discovered

Unseen

Queue: C → F → H → I → M → K

Visited: A, B



BFS Traversal

Step 6: Process node C

Legend

Complete

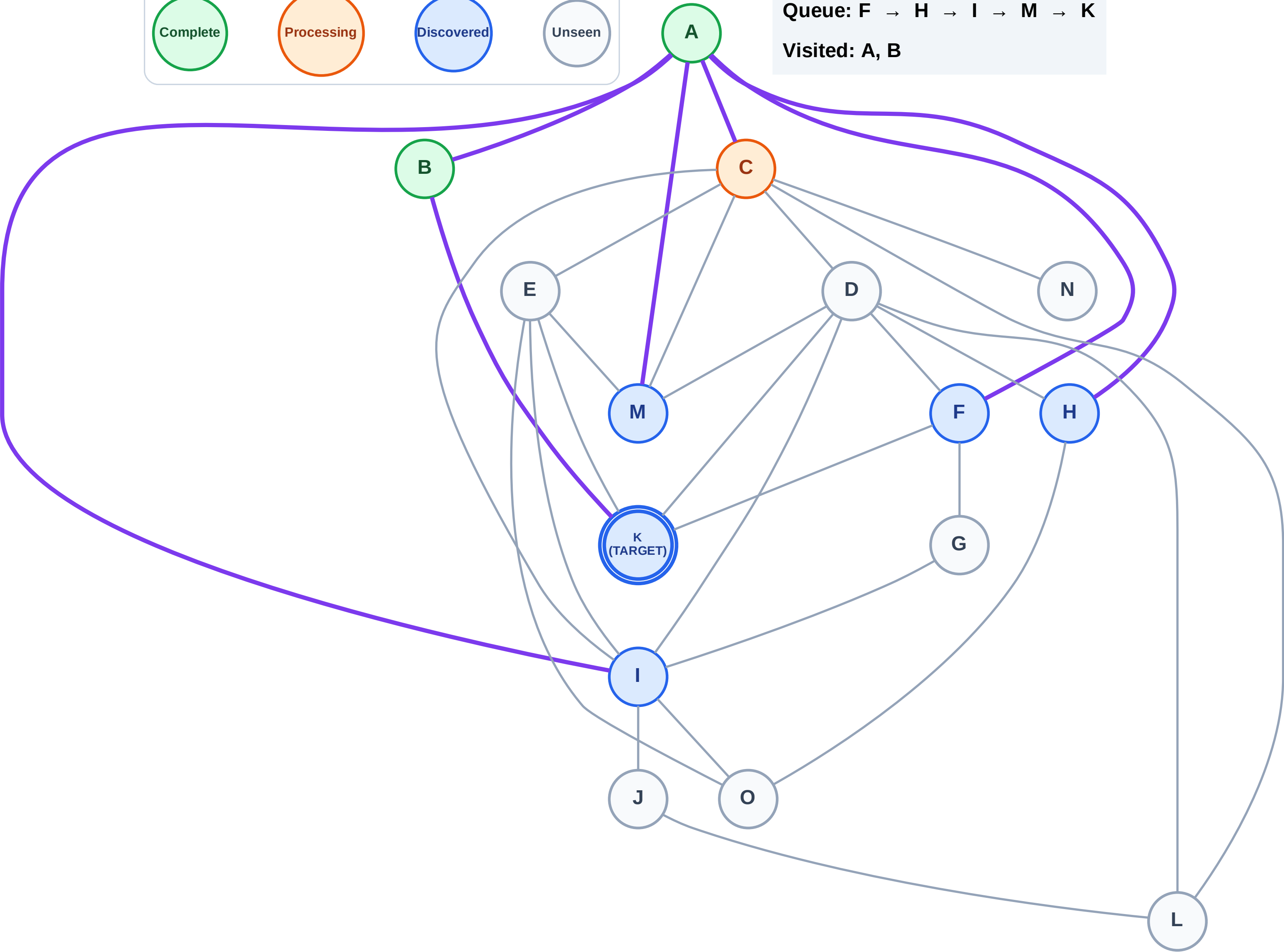
Processing

Discovered

Unseen

Queue: F → H → I → M → K

Visited: A, B



BFS Traversal

Step 7: Discover D, E, L, N from C

Legend

Complete

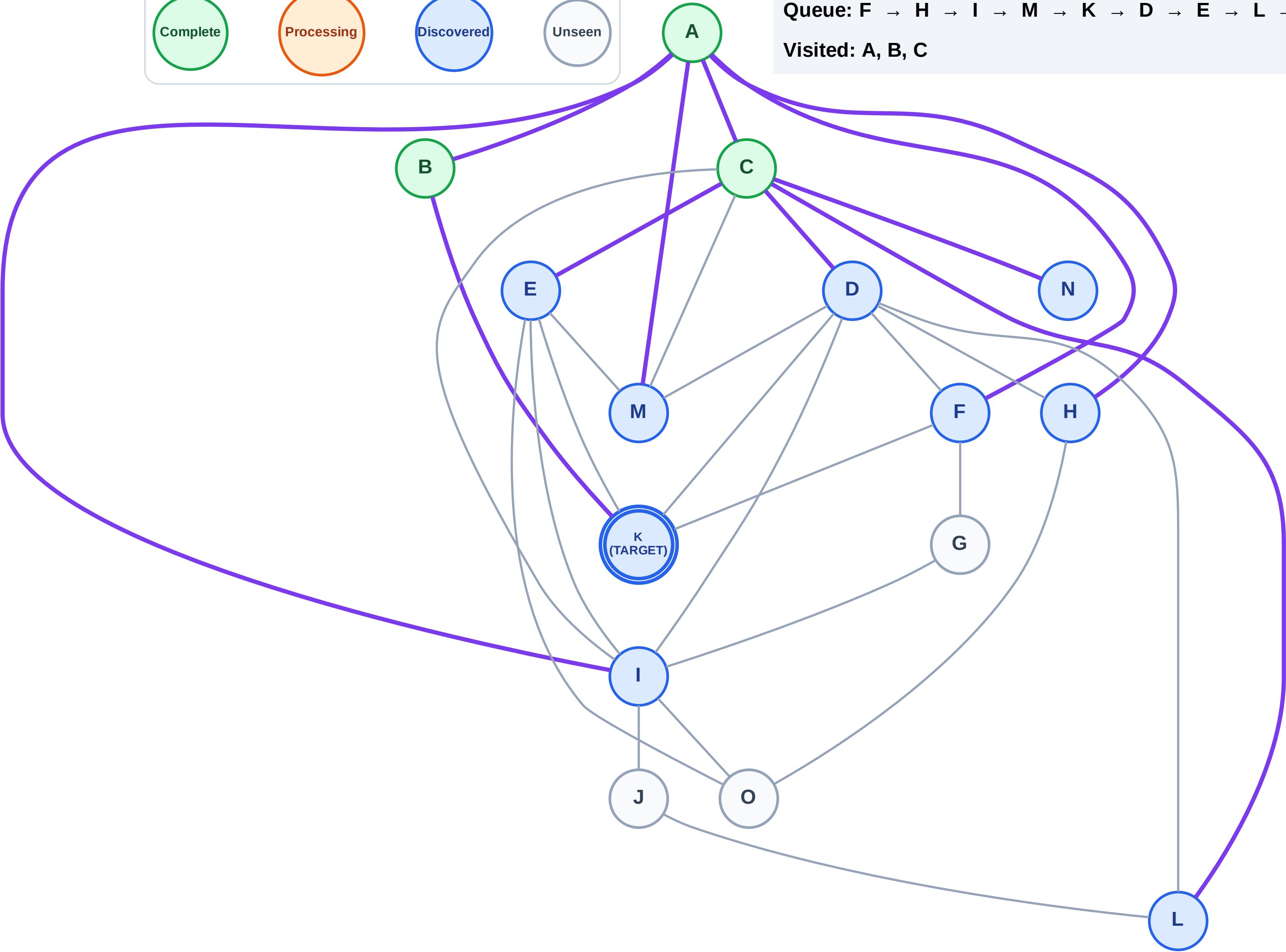
Processing

Discovered

Unseen

Queue: F → H → I → M → K → D → E → L → N

Visited: A, B, C



BFS Traversal

Step 8: Process node F

Legend

Complete

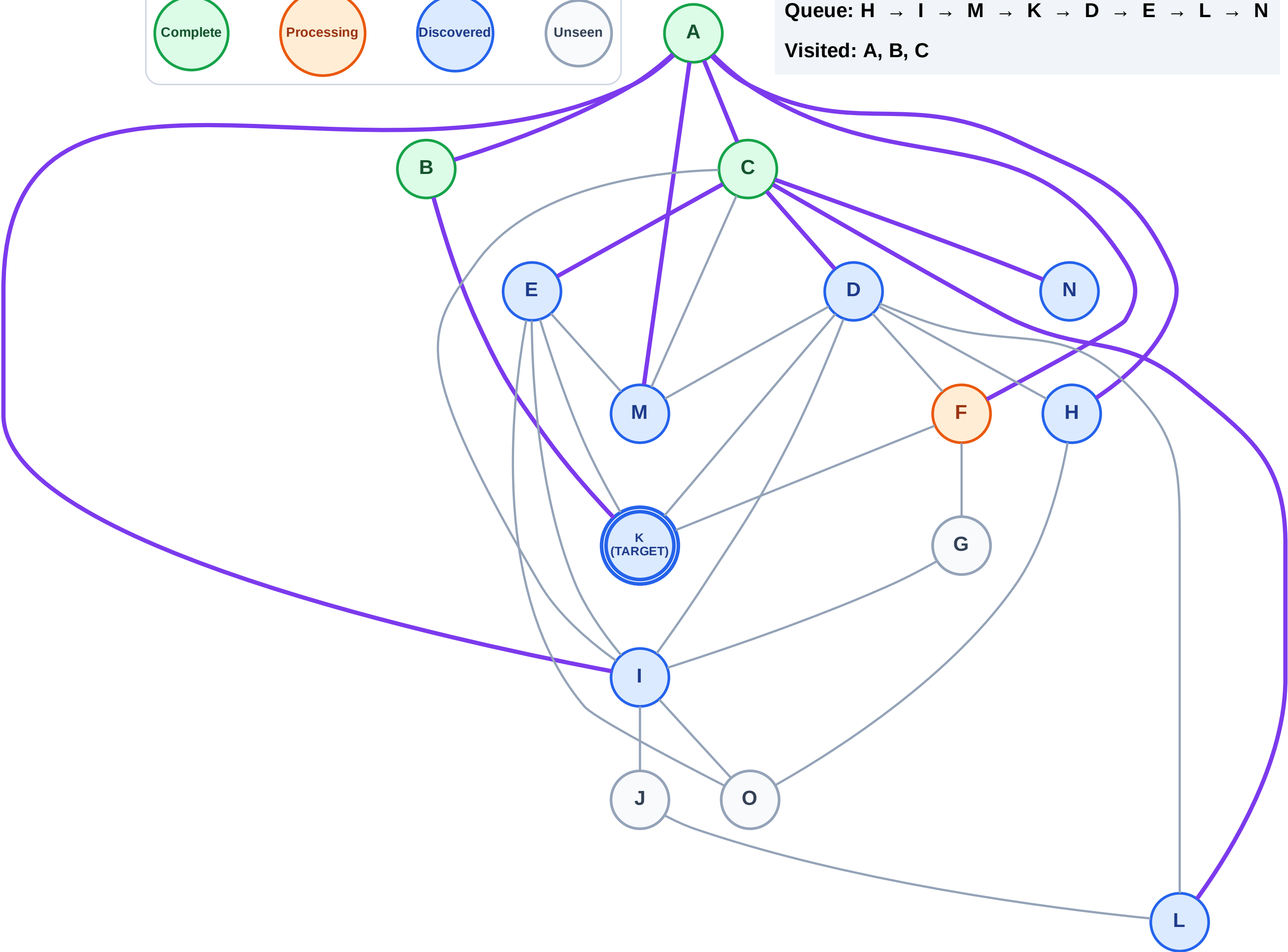
Processing

Discovered

Unseen

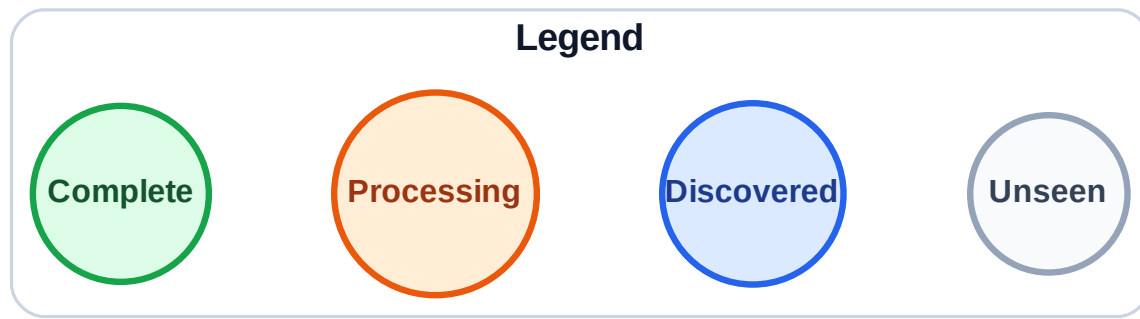
Queue: H → I → M → K → D → E → L → N

Visited: A, B, C



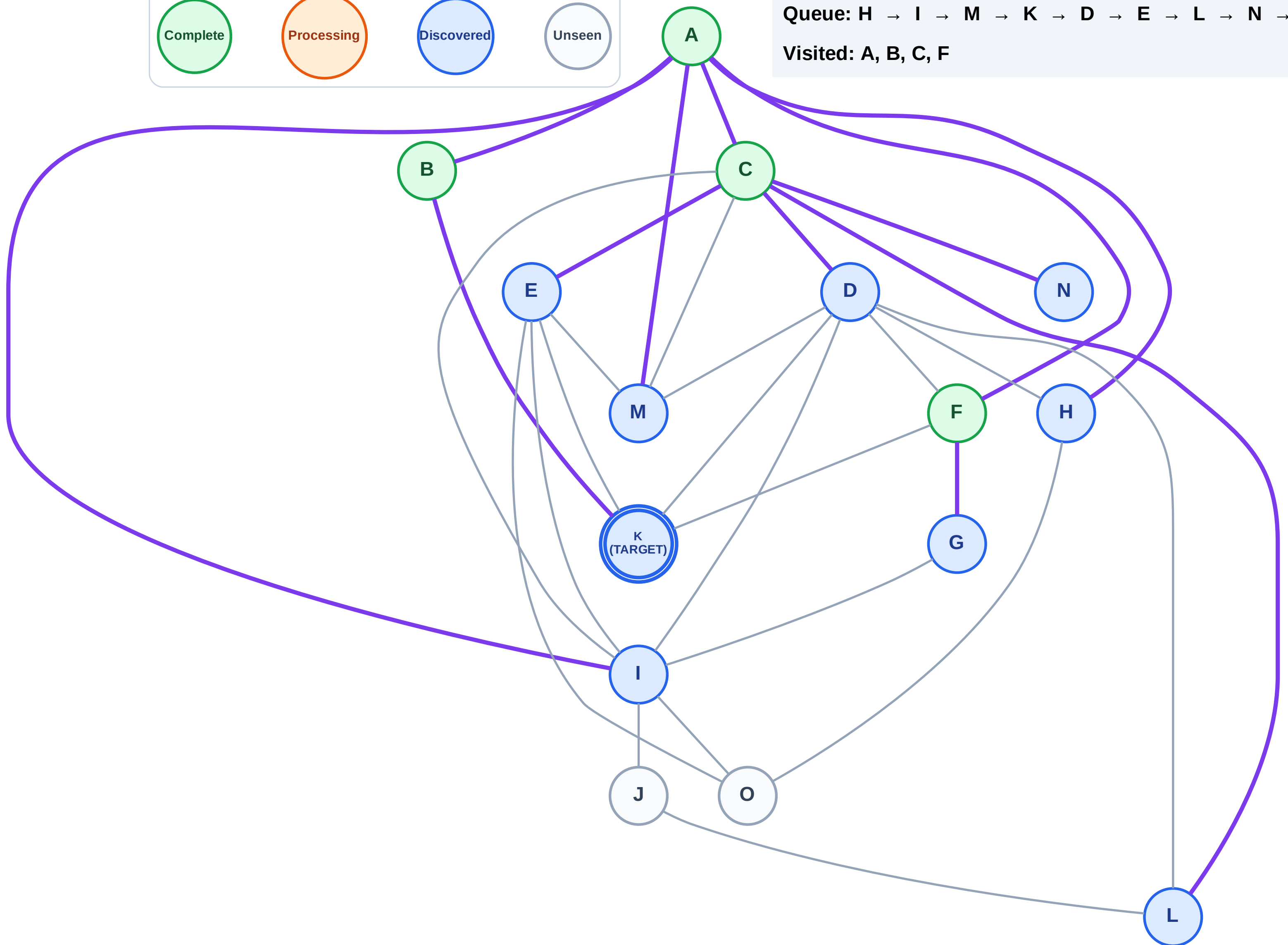
BFS Traversal

Step 9: Discover G from F



Queue: H → I → M → K → D → E → L → N → G

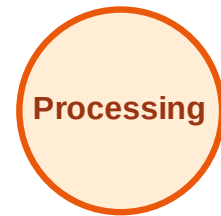
Visited: A, B, C, F



BFS Traversal

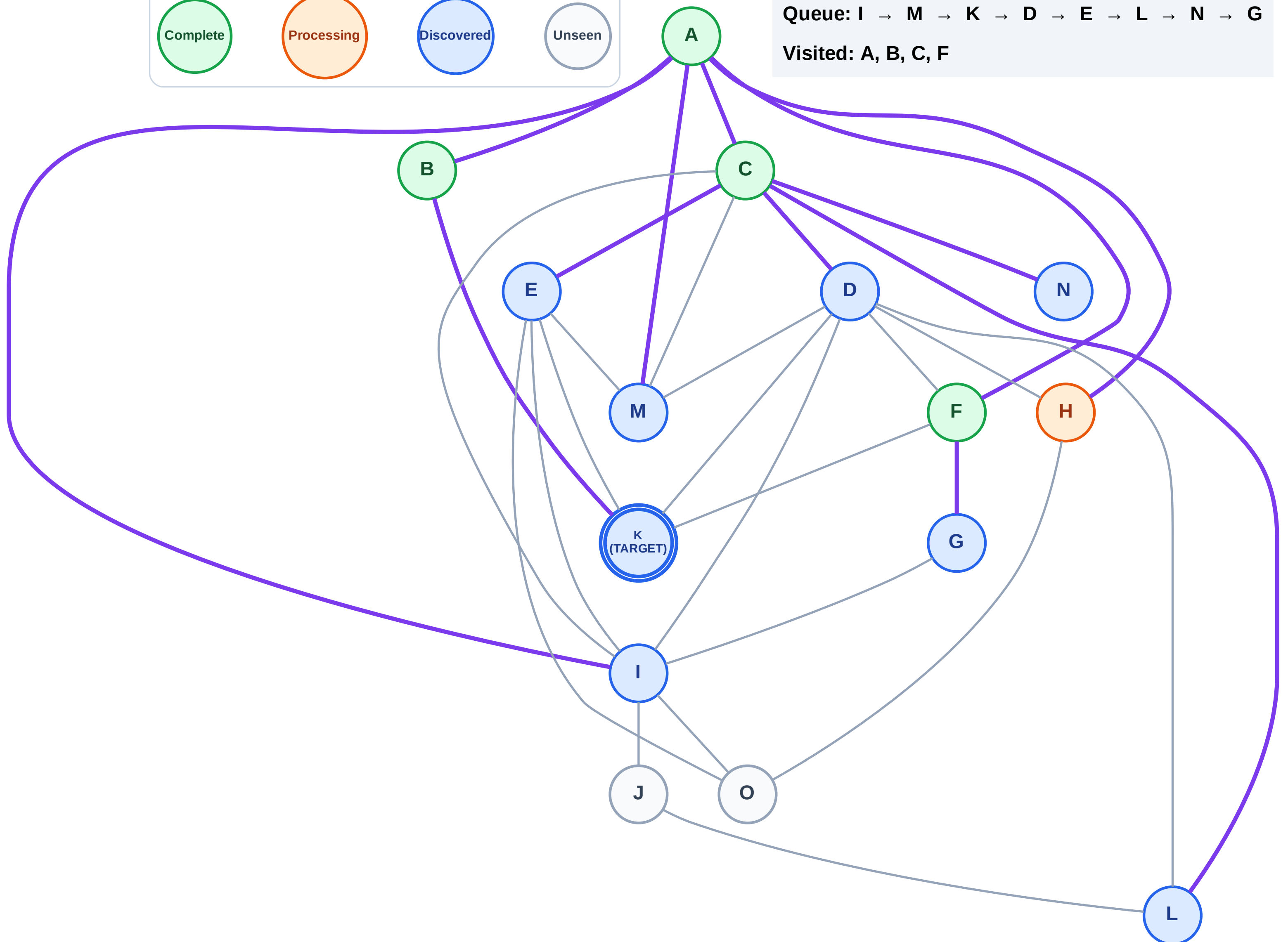
Step 10: Process node H

Legend



Queue: I → M → K → D → E → L → N → G

Visited: A, B, C, F



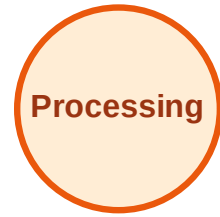
BFS Traversal

Step 11: Discover O from H

Legend



Complete



Processing



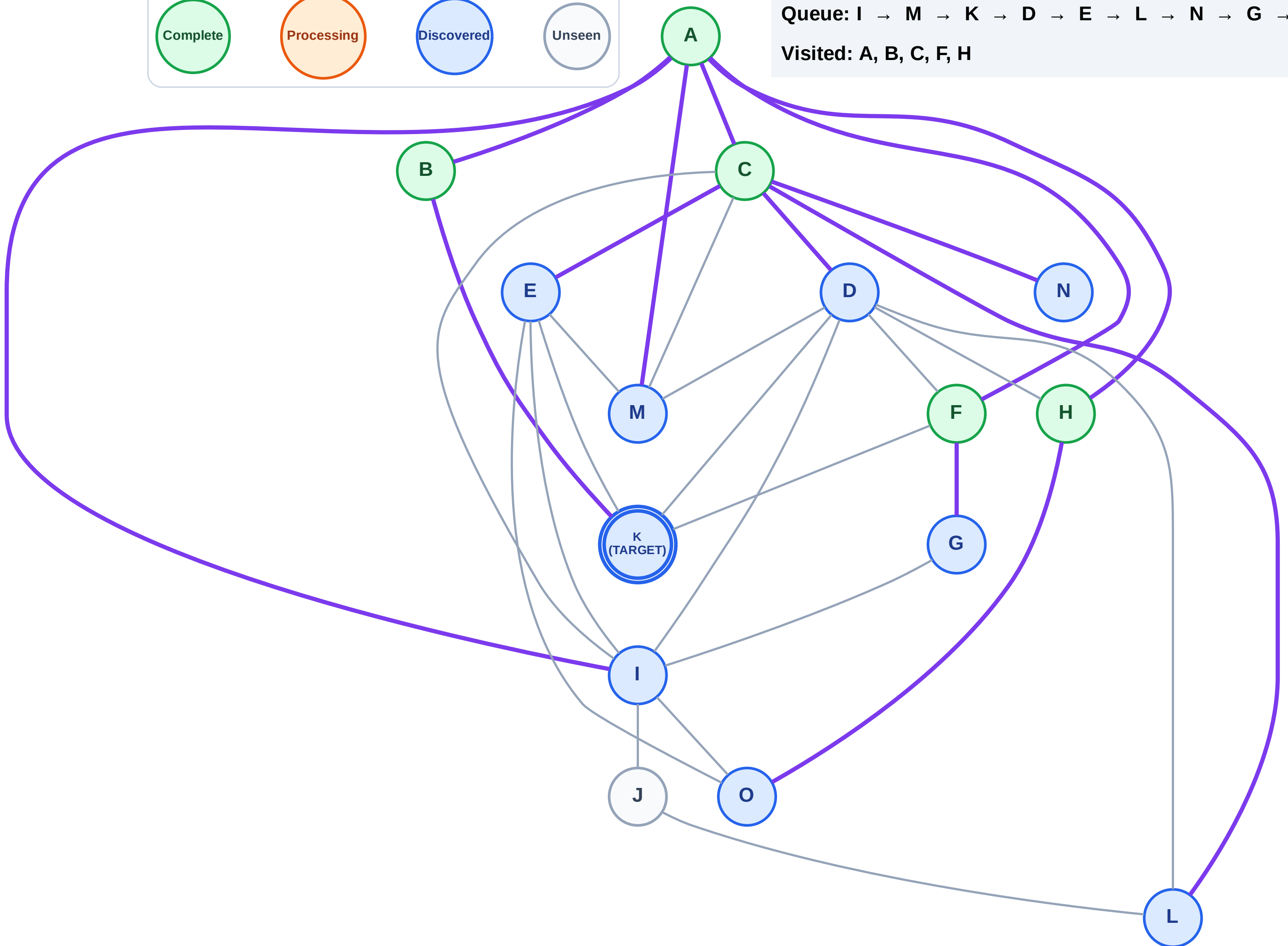
Discovered



Unseen

Queue: I → M → K → D → E → L → N → G → O

Visited: A, B, C, F, H



BFS Traversal

Step 12: Process node I

Legend

Complete

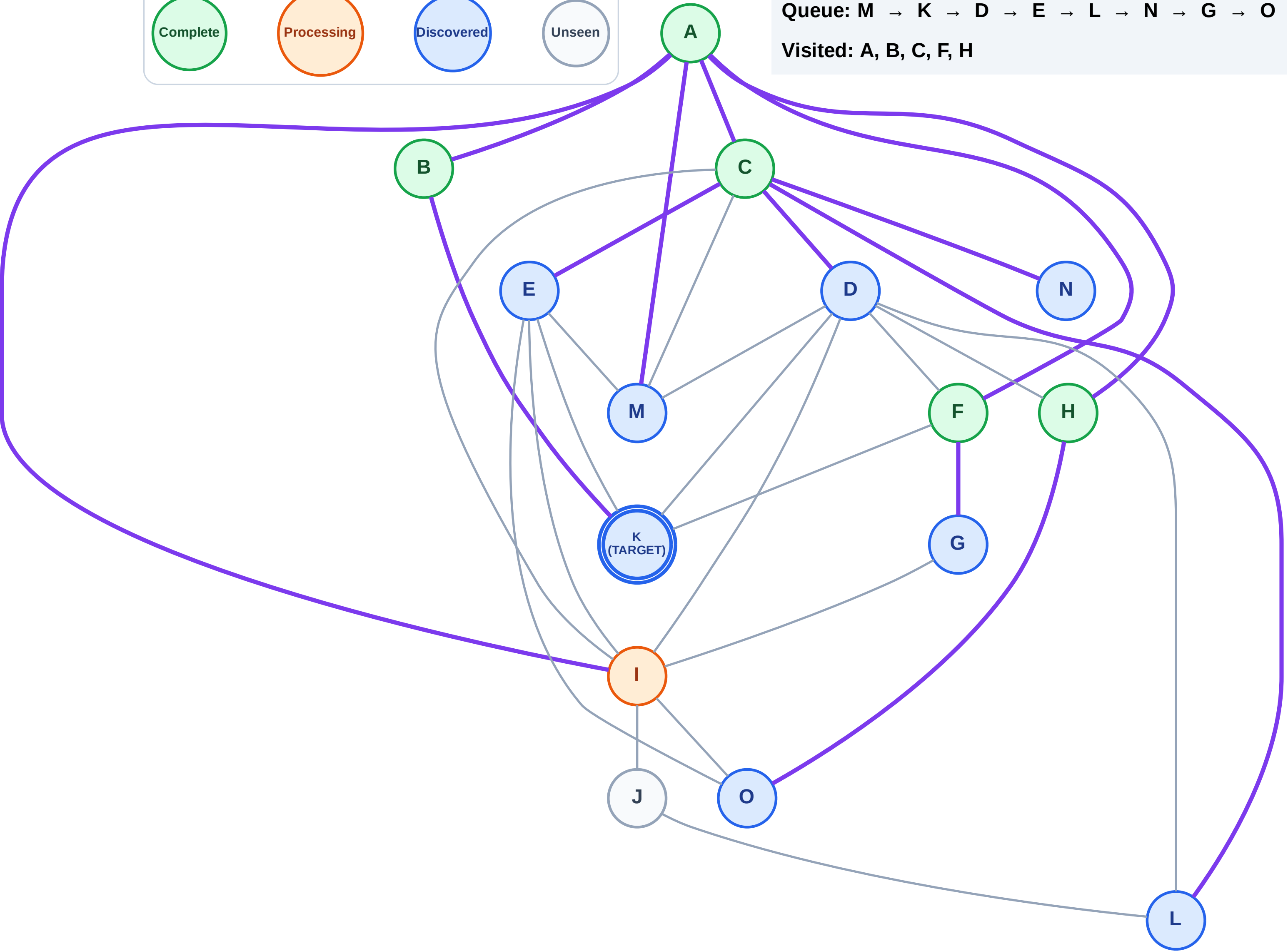
Processing

Discovered

Unseen

Queue: M → K → D → E → L → N → G → O

Visited: A, B, C, F, H



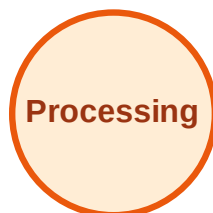
BFS Traversal

Step 13: Discover J from I

Legend



Complete



Processing



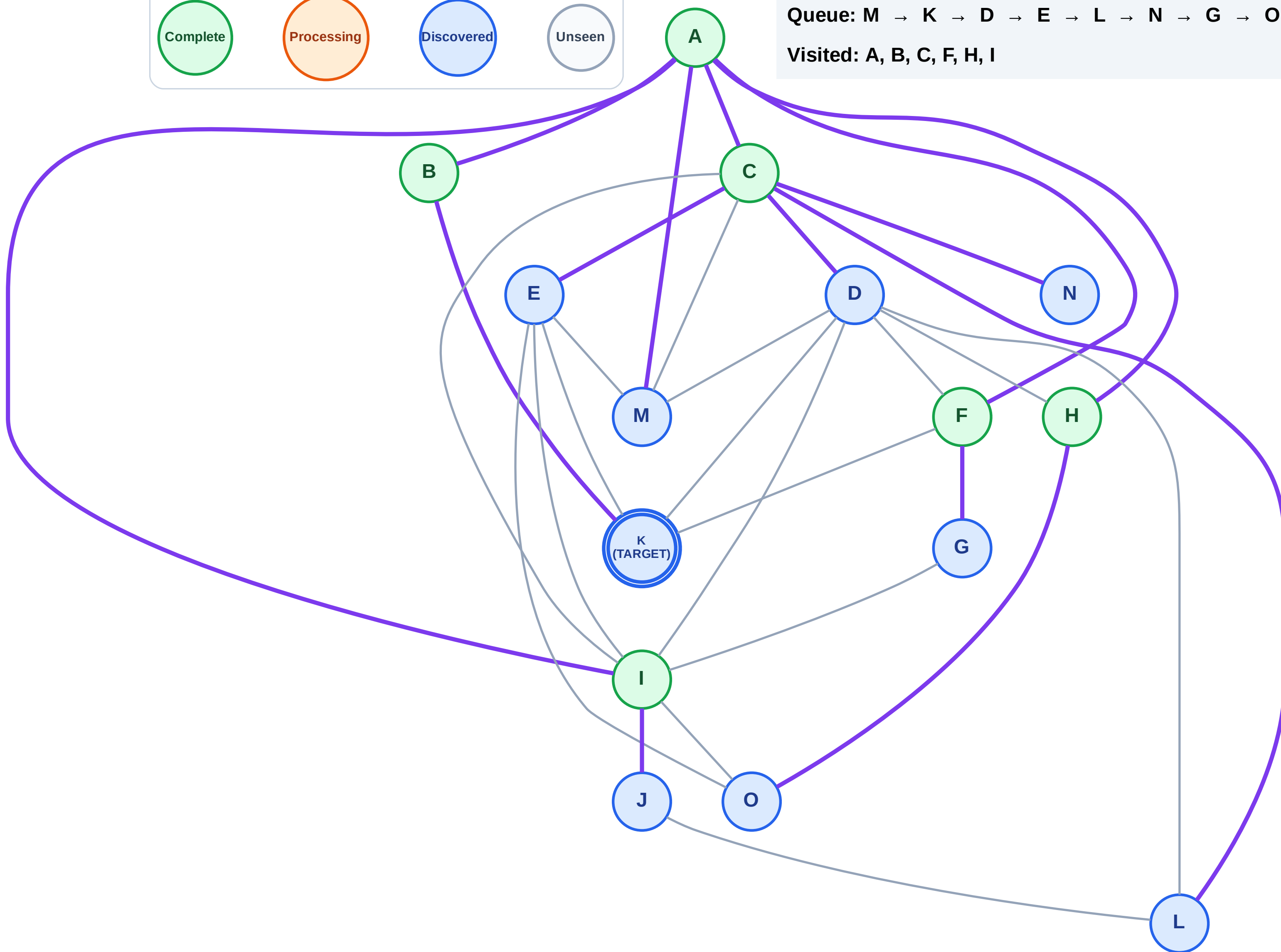
Discovered



Unseen

Queue: M → K → D → E → L → N → G → O → J

Visited: A, B, C, F, H, I



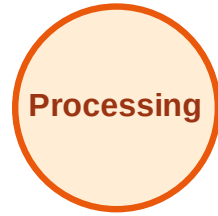
BFS Traversal

Step 14: Process node M

Legend



Complete



Processing



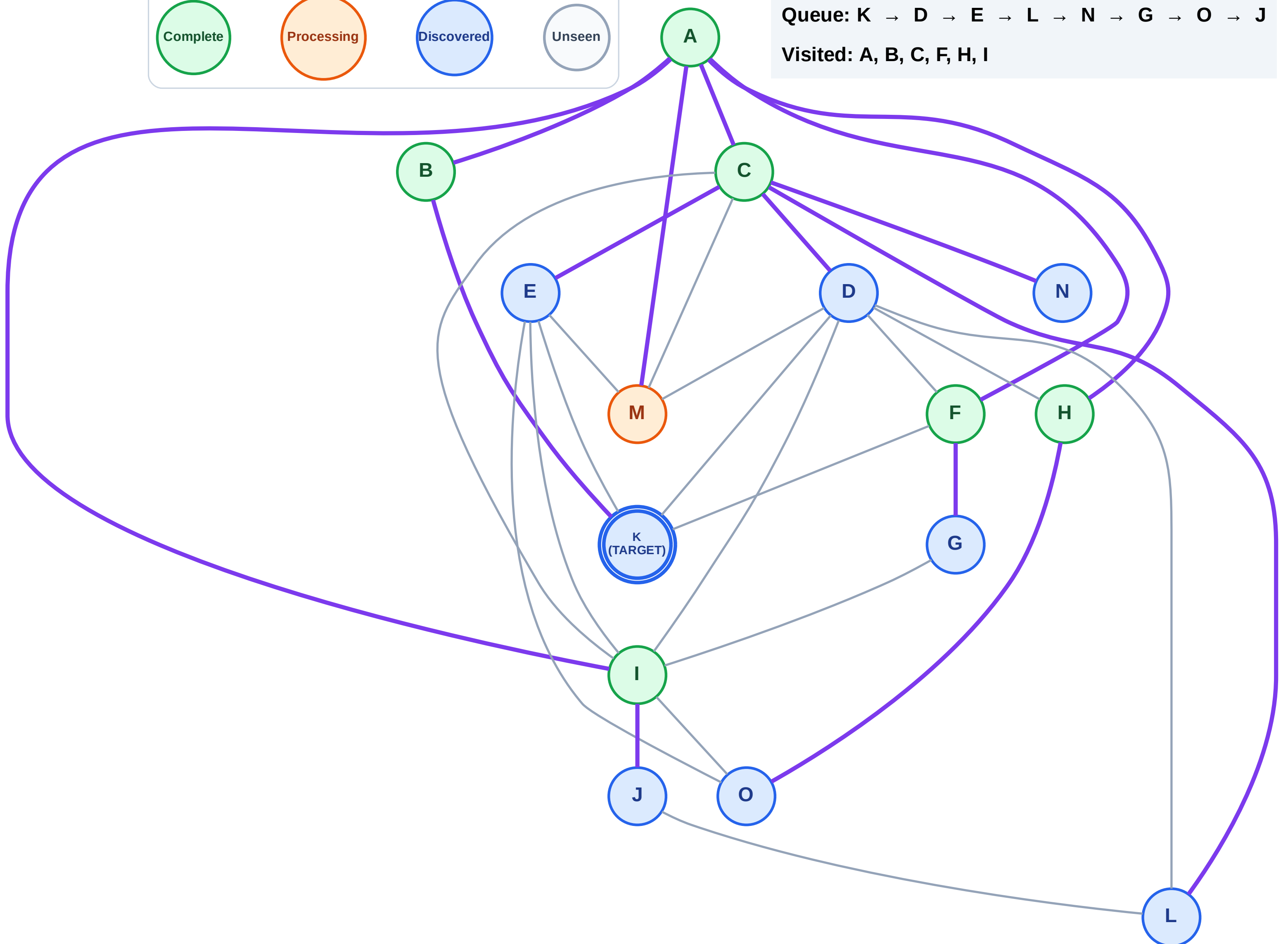
Discovered



Unseen

Queue: K → D → E → L → N → G → O → J

Visited: A, B, C, F, H, I



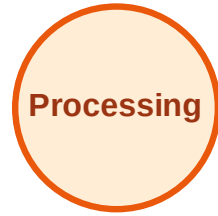
BFS Traversal

Step 15: No new neighbors from M

Legend



Complete



Processing



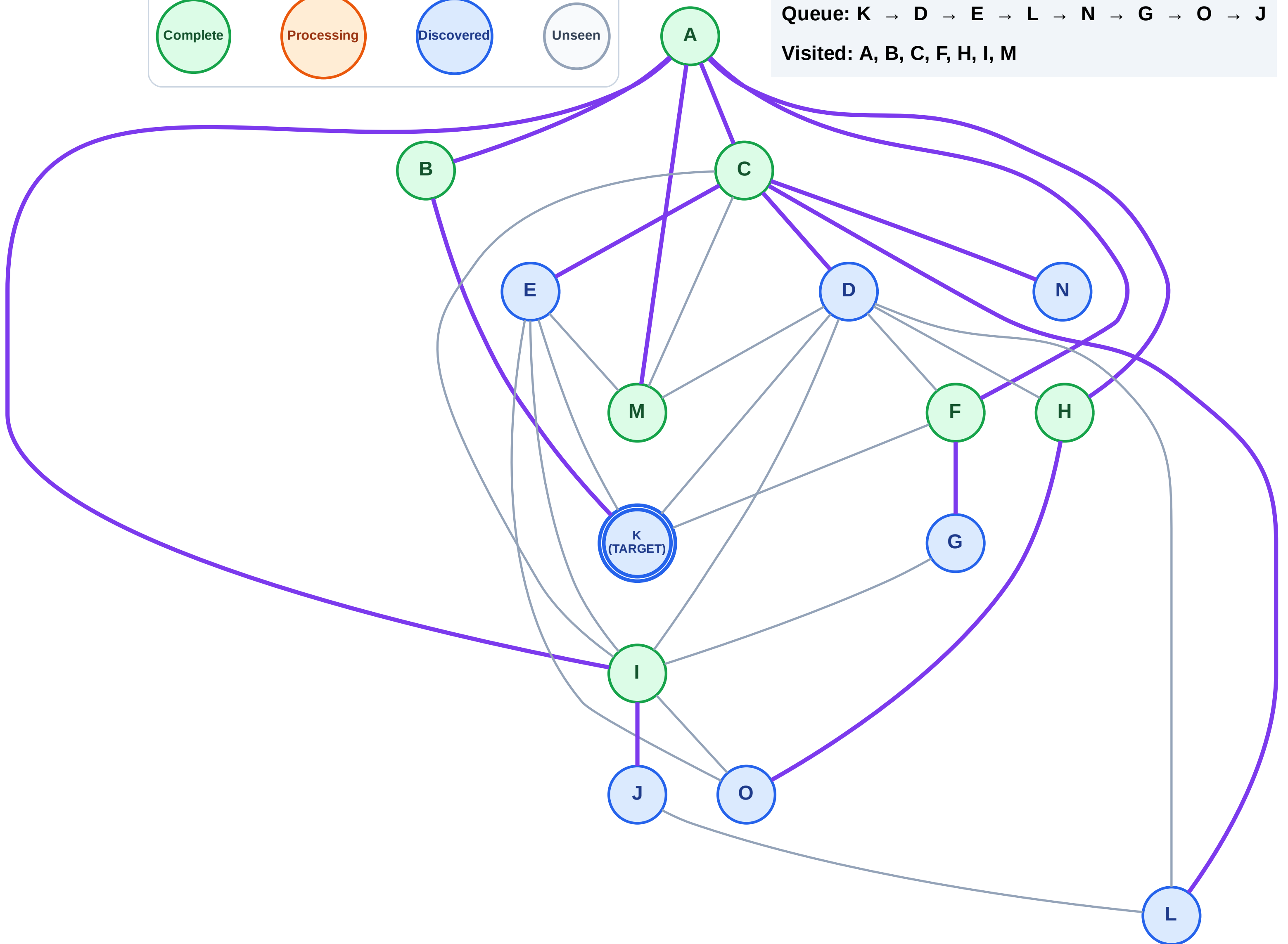
Discovered



Unseen

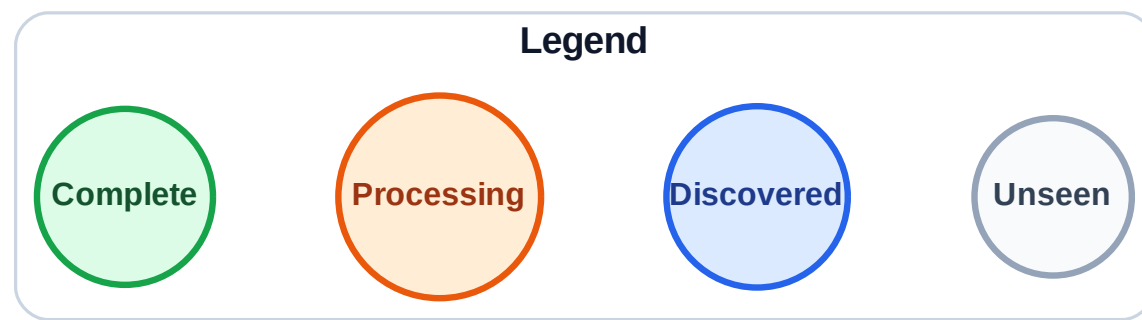
Queue: K → D → E → L → N → G → O → J

Visited: A, B, C, F, H, I, M



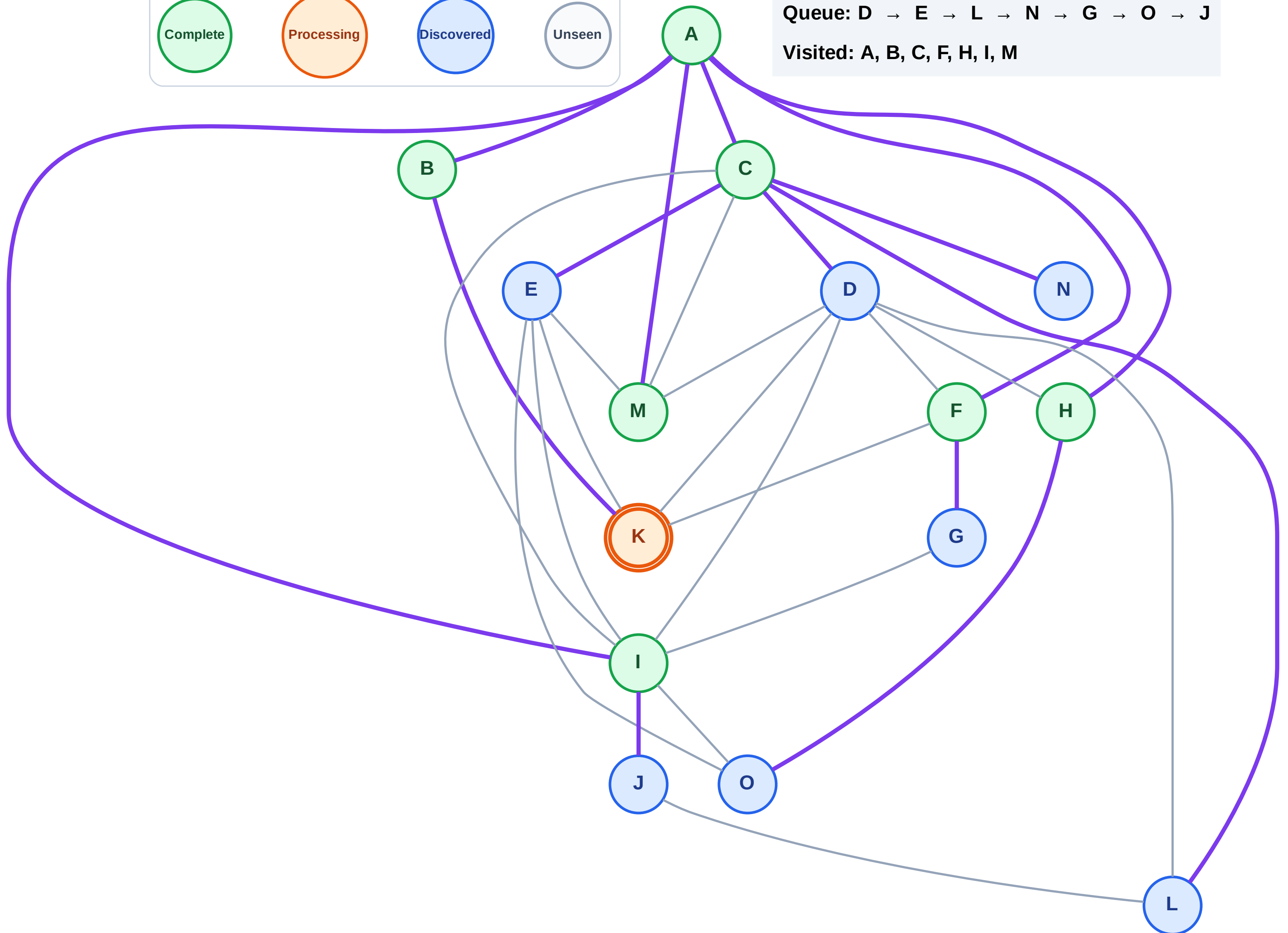
BFS Traversal

Step 16: Process node K



Queue: D → E → L → N → G → O → J

Visited: A, B, C, F, H, I, M



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

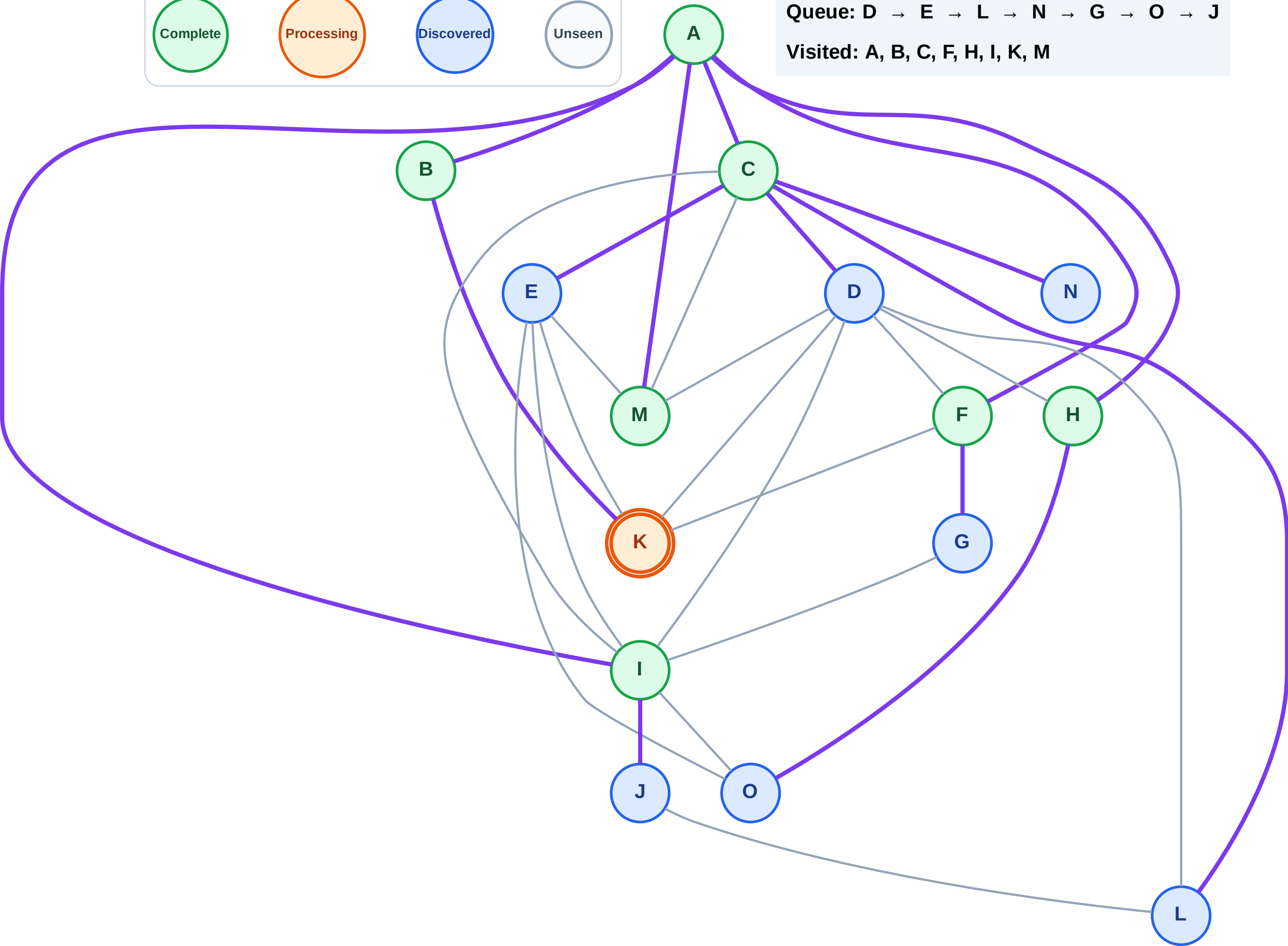
Processing

Discovered

Unseen

Queue: D → E → L → N → G → O → J

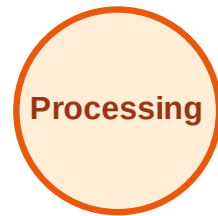
Visited: A, B, C, F, H, I, K, M



DFS Traversal

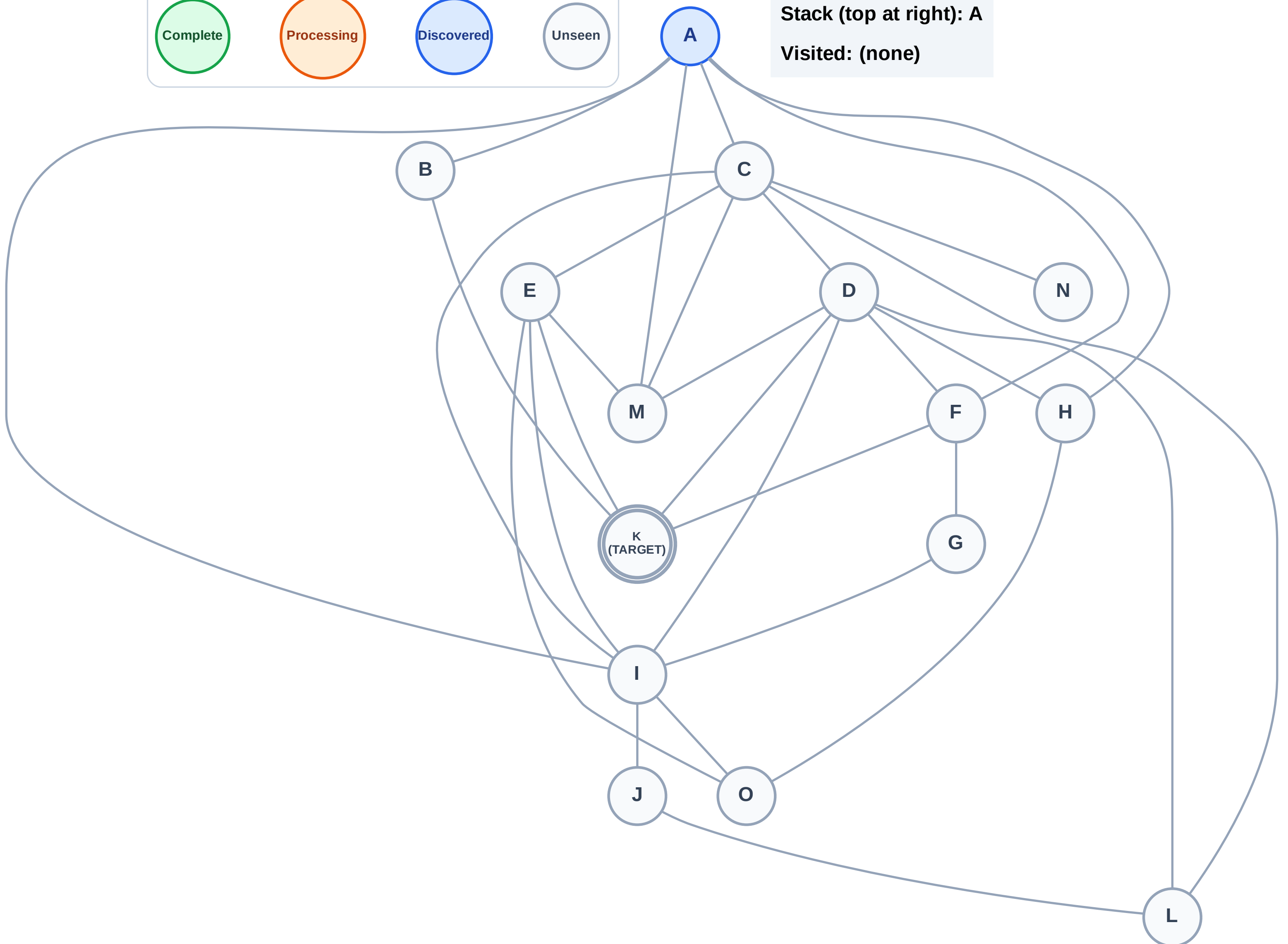
Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A

Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

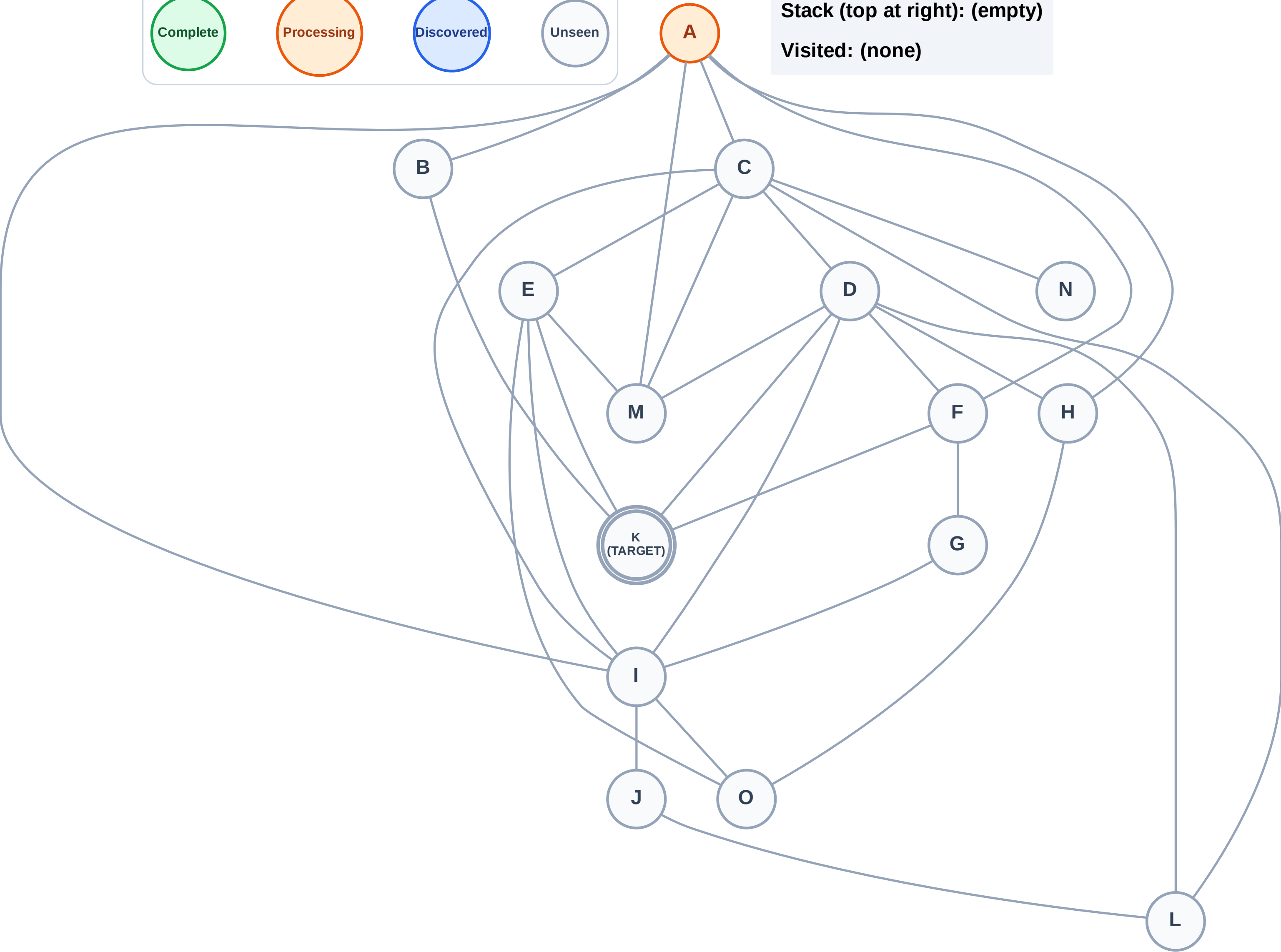
Complete

Processing

Discovered

Unseen

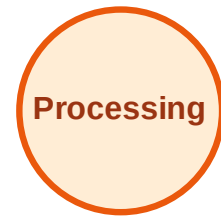
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

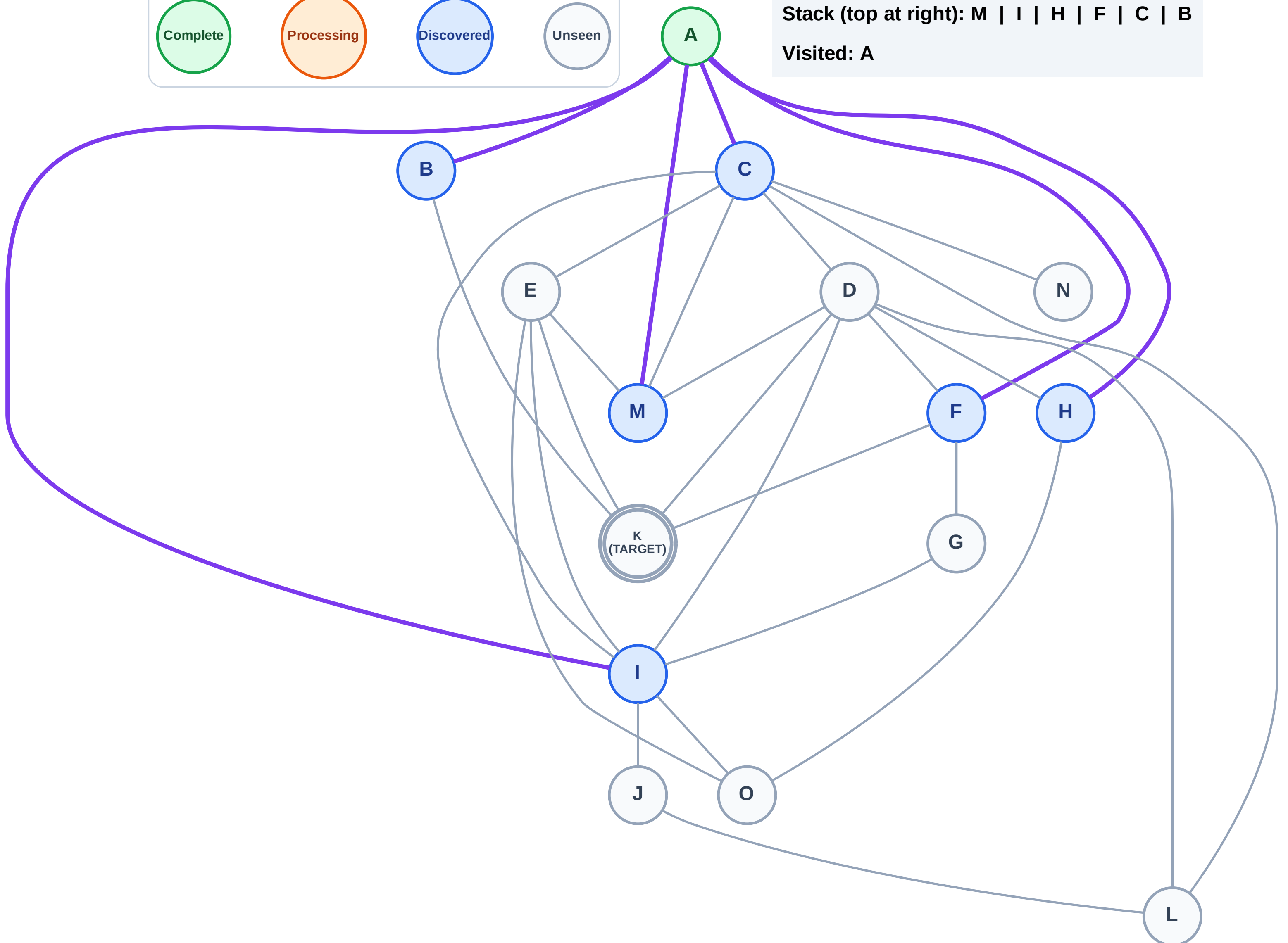
Step 3: Discover M, I, H, F, C, B from A

Legend



Stack (top at right): M | I | H | F | C | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

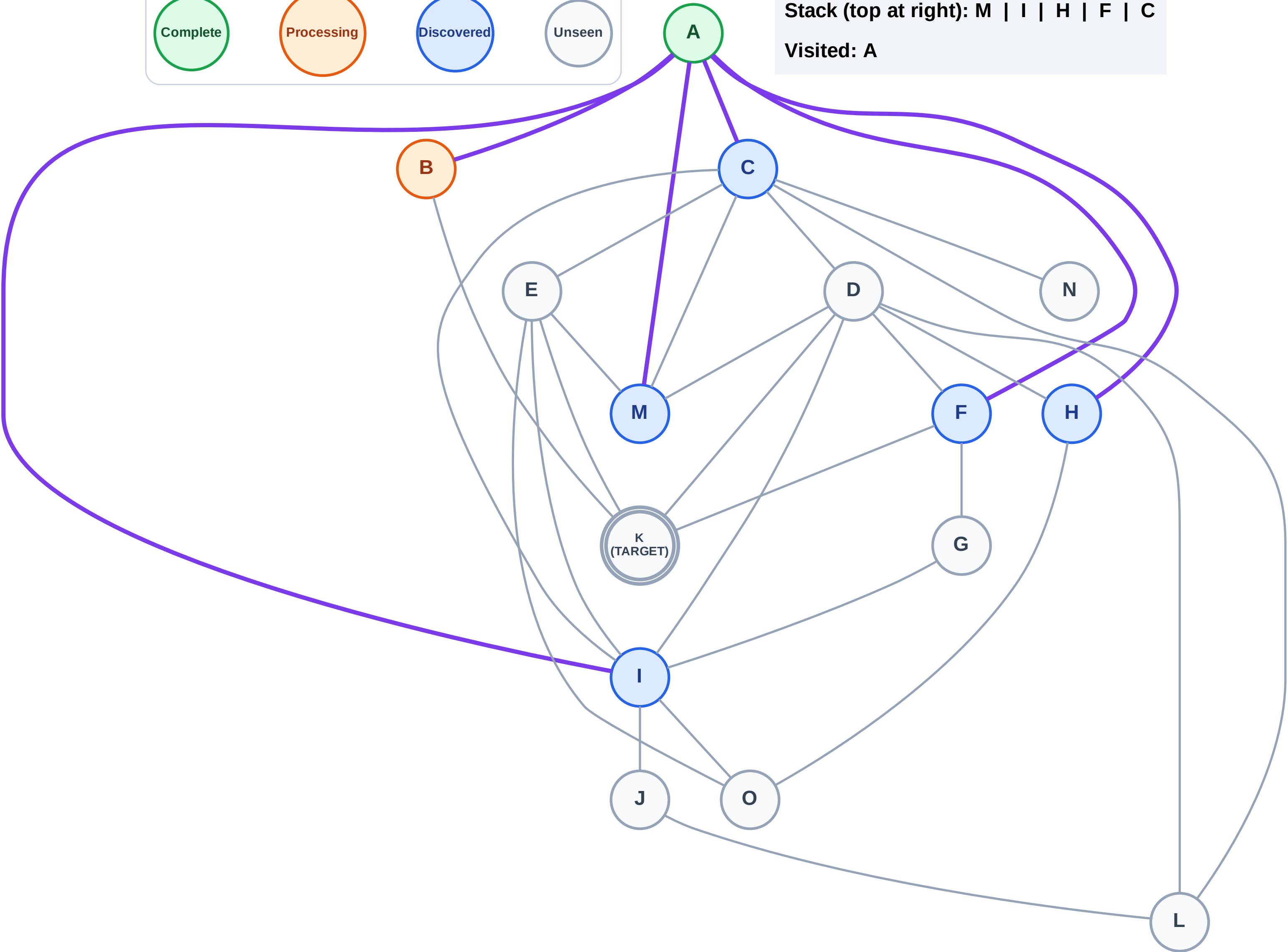
Processing

Discovered

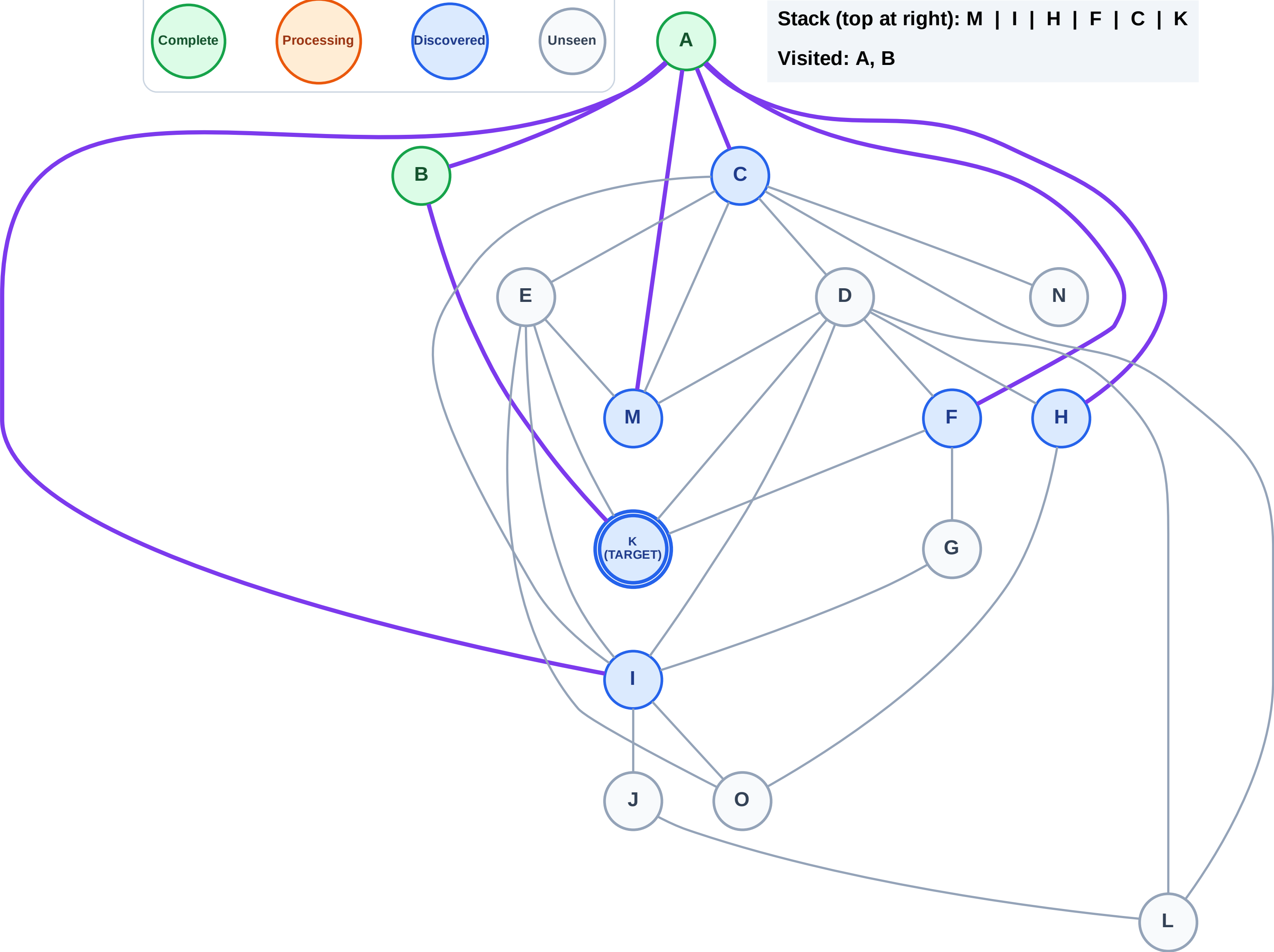
Unseen

Stack (top at right): M | I | H | F | C

Visited: A



Step 5: Discover K from E



DFS Traversal

Step 6: Process node K

Legend

Complete

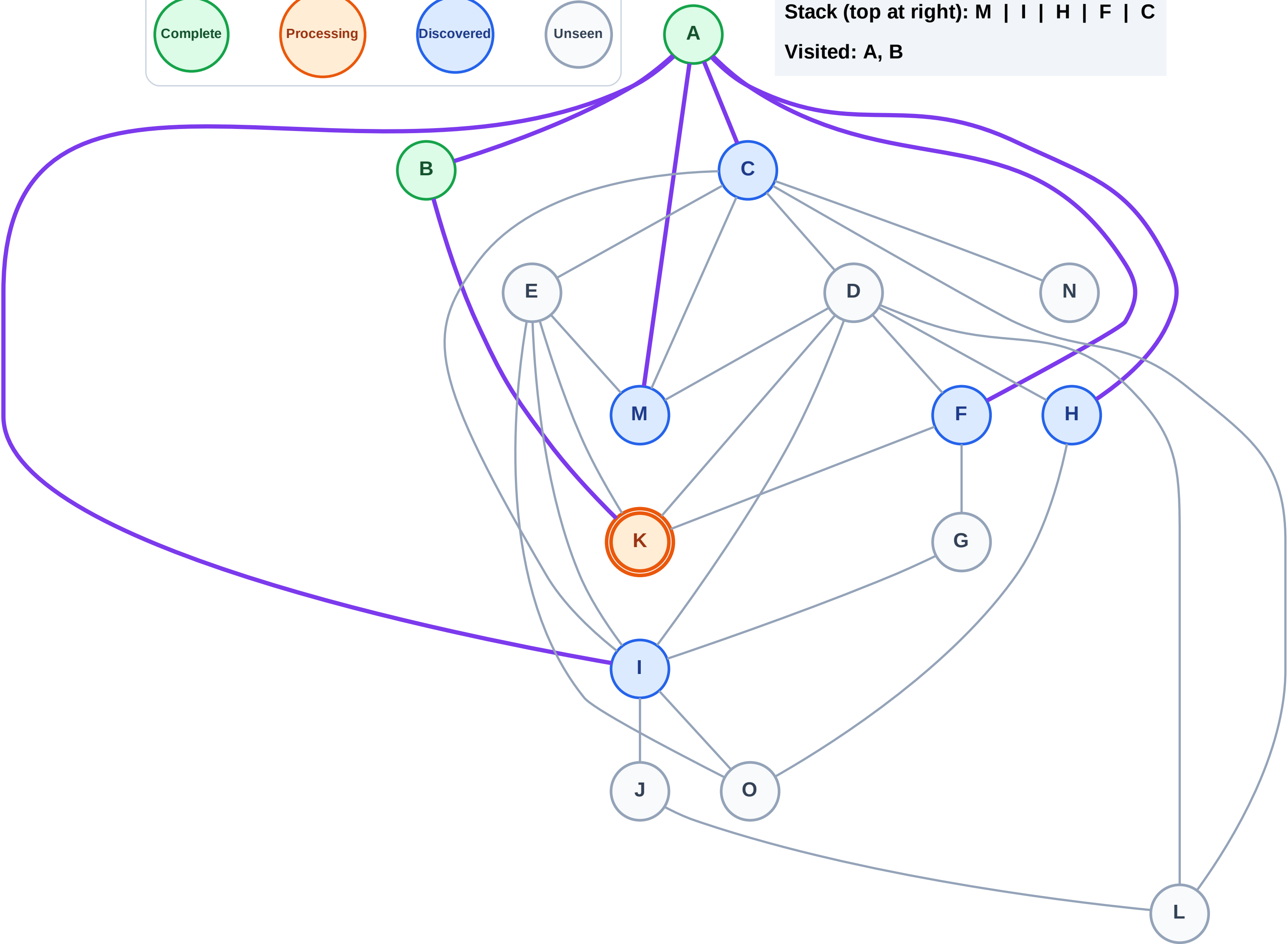
Processing

Discovered

Unseen

Stack (top at right): M | I | H | F | C

Visited: A, B



DFS Traversal

Step 7: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

Processing

Discovered

Unseen

Stack (top at right): M | I | H | F | C
Visited: A, B, K

